

Optics Letters

Additivity of the coefficient of thermal expansion in silicate optical fibers

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A model that predicts the material additivity of the thermal expansion coefficient in the binary silicate glasses most commonly used for present (GeO₂-SiO₂, P₂O₅-SiO₂, B₂O₃-SiO₂, and Al₂O₃-SiO₂) and emerging (BaO-SiO₂) optical fibers is proposed. This model is based on a derivation of the expression for the coefficient of thermal expansion in isotropic solids, and gives direct insight on the parameters that govern its additivity in silicate glasses. Furthermore, a consideration of the local structural environment of the glass system is necessary to fully describe its additivity behavior in the investigated systems. This Letter is important for better characterizing and understanding the impact of temperature and internal stresses on the behavior of optical fibers. © 2017 Optical Society of America

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Optical fibers are ubiquitous tools for global communications and, despite all the effort that has gone into their development, the accurate modeling and deduction of selected properties such as the linear coefficient of thermal expansion (α) lag behind those of other properties such as the linear refractive index. The thermal expansion coefficient is particularly important since, for example, its use in design has led to the realization of several practical types of polarization-maintaining fibers [1]. On the other hand, α can also lead to parasitic stress-optic effects [2], and it influences other fiber properties such as the thermo-optic coefficient, dn/dT [3], which can strongly impact modal performance and lead to mode instabilities in large mode area (LMA) fiber lasers [4,5]. Therefore, understanding and better predicting the coefficient of thermal expansion of these optical systems is crucial in the design of next-generation fibers.

Several models have been proposed over the past century to determine the material additivity of α based on empirical

formulae. For example, and among others, Winkelmann and Schott [6], English and Turner [7], and Hall [8] used empirical linear coefficients as a function of oxide percent of the glass, whereas Gilard and Dubrul [9] used a “parabolic equation,” where α additivity is described by a second order polynomial function $ax + bx^2$ where x is the weight percent of oxide and a and b are fit parameters. More recently, Doweidar [10] developed an additive model where α is a function of the structural units per that component in the glass. In this Letter, we propose a new additive model of the coefficient of thermal expansion in silicate glasses based on its derivation in isotropic solids such as glasses, and including the notion of structural units from Ref. [10]. The model is developed for additives into SiO₂ for the canonical dopants (i.e., GeO₂, P₂O₅, Al₂O₃, and B₂O₃), along with an emerging network modifier dopant (i.e., BaO) which is gaining attention for the developing of intrinsically low Brillouin gain fibers [11]. The novelty of this Letter is to show that the evolution of α in different ubiquitous silicate systems scales with the density variation induced by the volume fraction of the structural units that form the glass network.

The additivity of density (ρ) in silicate glasses can be described as a function of the volume fraction of each individual component (e.g., SiO₂ or GeO₂) from which the glass is made [12–14]. First, the density can be expressed in its most trivial form as a function of mass (m) and volume (V):

$$\rho = \frac{m}{V}. \quad (1)$$

Then, it is assumed that, in a multicomponent glass, each individual constituent “ i ” will additively contribute to the total molar mass and volume. Therefore, the aggregate value ρ of a silicate glass system can be written as follows:

$$\rho = \sum_i \frac{m_i}{V_i} = \sum_i \frac{\rho_i V_i}{V_i} = \sum_i \rho_i x_i \quad (2)$$

where ρ_i and x_i are the density and the volume fraction of each constituent of the glass. In a binary system composed of A and B , x_A would take the following form:

$$x_A = \frac{\frac{[A]M_A}{\rho_A}}{\frac{[A]M_A}{\rho_A} + \frac{[B]M_B}{\rho_B}}. \quad (3)$$

Here, $[A]$ and $[B]$ are the molar percent of components A and B , and M_A , M_B , ρ_A , and ρ_B are their respective molar masses and densities. Now, by differentiating ρ with respect to the temperature T , we obtain

$$\frac{\partial \rho}{\partial T} = \sum_i \frac{\partial \rho_i x_i}{\partial T}. \quad (4)$$

Next, we recall the expression of the relative derivative of the volume (here expressed in terms of density) for isotropic solids:

$$\beta = -\frac{1}{\rho} \frac{\partial \rho}{\partial T}, \quad (5)$$

with β being the volumetric coefficient of thermal expansion. From Eq. (4), and assuming $\beta \approx 3\alpha$ and substituting the temperature derivatives via Eq. (5), we finally arrive at the following equation:

$$\alpha = \sum_i \frac{\alpha_i \rho_i x_i}{\rho}. \quad (6)$$

With Eq. (6), a simple expression for the linear coefficient of the thermal expansion in glasses based on the additivity argument of ρ as enunciated earlier is derived.

As noted above, a series of GeO_2 - SiO_2 , P_2O_5 - SiO_2 , B_2O_3 - SiO_2 , Al_2O_3 - SiO_2 , and BaO - SiO_2 binary glasses was chosen to validate this additivity approach. The necessary material properties to perform the calculation of the additivity are summarized in Table 1. The data are fit using Eq. (6), with α for fused SiO_2 taken as $5.5 \times 10^{-7} \text{ K}^{-1}$ for all systems investigated [18]. Thus, the value of α for the non-silica compound is taken as a fit parameter. The data, along with graphical fits using Eq. (6), are displayed in Fig. 1. Since the molar volume of each individual constituent is employed in the Eq. (6) calculation, the x axis of Fig. 1 is expressed in mole percent. It can clearly be seen that Eq. (6) works very well in the cases of the GeO_2 - SiO_2 , P_2O_5 - SiO_2 , and BaO - SiO_2 systems. However, it does not fit the data in the Al_2O_3 - SiO_2 and B_2O_3 - SiO_2 systems.

This suggests that Eq. (6) is not sufficient to entirely describe the behavior of the coefficient of thermal expansion, and other factors must account for the additivity. As suggested in [10,26], a glass can be considered as the sum of structural units that form the glass network and, therefore, the coefficient of thermal expansion of a glass must be related to the relative proportion of a structural unit that each mole of constituent occupies in the glass. For instance, in the GeO_2 - SiO_2 system, a SiO_4 tetrahedron is substituted by a GeO_4 tetrahedron. In the

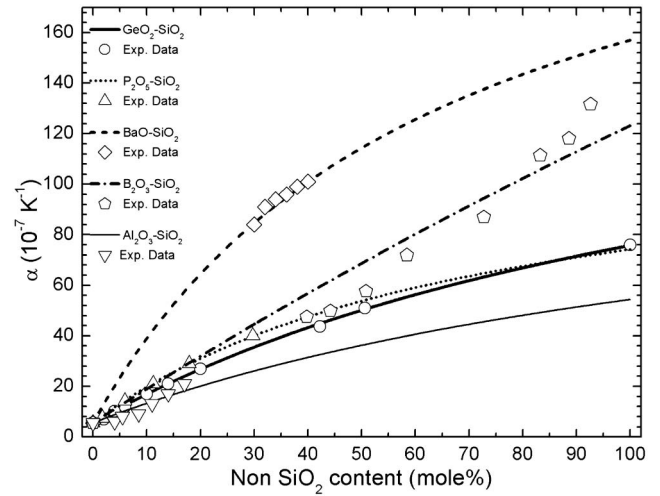


Fig. 1. Linear coefficient of thermal expansion (α) for some typical binary silicate systems as a function of non-silica content. The fitting of these data is performed with Eq. (6). Data were found in [19,20] for GeO_2 - SiO_2 , in [21] for B_2O_3 - SiO_2 and BaO - SiO_2 , citing [22] and [23], respectively, [24] for P_2O_5 - SiO_2 , and [25] for Al_2O_3 - SiO_2 . All the systems together are valid within a common range of 25°C – 100°C .

case of the P_2O_5 - SiO_2 system, the addition of phosphorous leads to the formation of single and double phosphorous centers in the silica network [27]. In Fig. 10 of the same reference, it is shown that as P_2O_5 content increases, there is an increase of double phosphorous centers, and a decrease of single phosphorous centers.

For this reason, we consider here P_2O_5 to progressively enter the silica network as double phosphorous centers, locally forming a “pair” of $\text{O} = \text{PO}_3$ tetrahedral units bonded with a common oxygen. In the case of B_2O_3 - SiO_2 , the substitution of one mole of SiO_2 for one mole of B_2O_3 will result in the substitution of one SiO_4 unit for two BO_3 units in the glass network. Additionally, we consider Al_2O_3 to enter the network by forming AlO_4 tetrahedral units. Therefore, to appropriately take these structural effects into consideration, Eq. (6) is modified to

$$\alpha = \sum_i \frac{\alpha_i \rho_i y_i}{\rho}, \quad (7)$$

where x_i is substituted by y_i , which is now defined as the volume fraction of the compound i per structural unit volume T_i of that component. For a binary system composed of A and B , Eq. (3) will become

$$y_A = \frac{\frac{[A]M_A}{\rho_A T_A}}{\frac{[A]M_A}{\rho_A T_A} + \frac{[B]M_B}{\rho_B T_B}}. \quad (8)$$

If we consider the GeO_2 - SiO_2 , P_2O_5 - SiO_2 , and BaO - SiO_2 binary systems, $T(\text{SiO}_2) = T(\text{GeO}_2) = T(\text{P}_2\text{O}_5) = T(\text{BaO}) = 1$. Again, the substitution of SiO_2 for P_2O_5 results in the formation of a “pair” of $\text{O} = \text{PO}_3$ units sharing a common oxygen. Therefore, $T(\text{P}_2\text{O}_5) = 1$, not 2, as the structural unit is “ $2 \times \text{O} = \text{PO}_3$.” However, for a B_2O_3 - SiO_2 system, $T(\text{B}_2\text{O}_3) = 2$ since one mole of B_2O_3 leads to two distinct (i.e., not connected) BO_3 structural units. For the same reason as B_2O_3 , $T(\text{Al}_2\text{O}_3) = 2$ as we consider Al_2O_3

Table 1. Molar Mass (M) and Density (ρ) Used for the Calculations

Materials	$M(\text{g} \cdot \text{mol}^{-1})$	$\rho(\text{kg} \cdot \text{m}^{-3})$
SiO_2	60.08	2200
GeO_2	104.61	3650 [15]
B_2O_3	69.62	1820 [16]
P_2O_5	141.94	2390 [17]
Al_2O_3	101.96	3350 [2]
BaO	153.33	4688 [11]

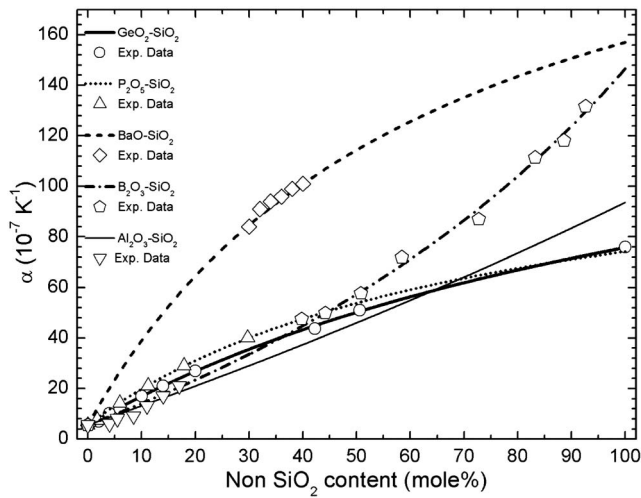


Fig. 2. Modified additivity prediction of the coefficient of thermal expansion (α) in binary silicate glasses using Eq. (7), instead of Eq. (6), and considering the additivity of α to be a function of the variation of density per structural glass unit.

incorporated into SiO_2 by forming independent AlO_4 units. Obviously, some limitations can arise from the difficulty to estimate the structural role that each dopant plays in the glassy network, as it is typically a strong function of glass composition. Nevertheless, using these assumptions, the data can be fitted using Eq. (7), instead of Eq. (6). The new fits are displayed in Fig. 2, and the correlation with the data is excellent.

Table 2 reports the determined values of the coefficient of thermal expansion for the glassy non-silica compounds using Eqs. (6) and (7), and are compared with data found in literature where appropriate.

Next, the utility in developing such a model is demonstrated with a practical example. As mentioned above, a novel BaO-doped optical fiber was previously developed [11] for applications requiring a reduced Brillouin gain coefficient. dn/dT for the BaO glass constituent was determined for one of the fabricated fibers to be $1.940 \times 10^{-5} \text{ K}^{-1}$ (Fiber C, Table II in [11]). This value is considerably larger than that of SiO_2 and other materials, suggesting that silicate fibers possessing baria may be more susceptible to thermal mode instabilities [4,5]. However, as is seen in Fig. 2, a barium silicate core fiber will

Table 2. Estimated Coefficient of Thermal Expansion of Amorphous Materials, Based on Additivity of Eqs. (6) and (7)

Materials	$\alpha(10^{-7} \text{ K}^{-1})$ Eq. (6)	$\alpha(10^{-7} \text{ K}^{-1})$ Eq. (7)	$\alpha(10^{-7} \text{ K}^{-1})$ Literature
SiO_2	5.5 ^a	5.5 ^a	5.5
GeO_2	75.7	75.7	75–80 [20,28]
B_2O_3	123.2	146.6	148 [29]
P_2O_5	74.2	74.2	70–100 [30] ^b
Al_2O_3	54.6	93.6	98.5 [2] ^c
BaO	157.0	157.0	—

^a $\alpha(\text{SiO}_2)$ is taken as $5.5 \times 10^{-7} \text{ K}^{-1}$ for the computation of each binary system.

^bExtrapolated from graph in Figs. 5 and 6.

^cDetermined by linear extrapolation of the same set of data used here.

have an α that can be much larger than that of a surrounding pure silica cladding. This effect was not taken into consideration in [11], and the dn/dT value listed above for BaO simply represents how the network itself is modified due to its presence in silica as a *binary* system, but also within a cladded geometry. Its value is essentially invalid for use in more complicated systems such as in ternary glasses [31], especially if another dopant brings about additional changes to α .

In principle, as the temperature of a fiber is increased, both the core and cladding will thermally expand. However, when $\alpha_{\text{core}} > \alpha_{\text{cladding}}$, the expansion of the core should be greater than that of the cladding. However, due to the presence of a much more voluminous and massive cladding surrounding it, thermal expansion of the core is restricted, and it appears as compressed relative to an unbound, freely expanding core. This pressure exerted on the core then also modifies the refractive index via the photoelastic effect [32,33]. This pressure causes a relative increase to the index of SiO_2 (a positive- p_{12} photoelasticity material) [34], whereas it results in a relative decrease in the index of BaO, since it is a negative- p_{12} material [11,35]. Using the procedure outlined in [36] and, by incorporating the model for α derived here, a global dn/dT value for the bulk BaO constituent can be determined.

In short, Eqs. (2) and (3) can be used to calculate the refractive index of a binary glass if the replacement $\rho \rightarrow n$ is made. To account for the α mismatch, it is shown [36] that n_i can be modified to

$$n = n_0 + \frac{dn}{dT}(T - T_0) + \frac{n_0^3}{2}(\alpha_{\text{core}} - \alpha_{\text{cladding}}) \times [2(p_{12} - \nu(p_{11} + p_{12})) + (p_{11} - 2\nu p_{12})](T - T_0), \quad (9)$$

where the subscript i denoting constituent has been left off for convenience. In Eq. (9), n_0 is the refractive index at a starting temperature (T_0 , e.g., room temperature), ν is the Poisson ratio, and the p 's are the usual Pockels' coefficients (all for the pure material). To determine dn/dT for the fiber, n is simply calculated as a function of T using Eqs. (2), (3), and (9). Armed with accurate models for α_{core} (the binary barium silicate glass, Fig. 2), dn/dT for BaO is the only unknown value in Eq. (9) (the remaining values may be found in [11]) and is used as a fitting parameter. Simply put, dn/dT for BaO is adjusted until the calculated fiber dn/dT matches measured data presented in [11]. A value of $1.270 \times 10^{-5} \text{ K}^{-1}$ is obtained, which is roughly 2/3 of that of the aggregated value. Again, the aggregated value takes the α mismatch between the core and cladding into consideration for the barium silicate *binary* system. The new, bulk BaO value may now be used in the design of more complex silicate systems, thus clearly demonstrating the importance of suitable models for the coefficient of thermal expansion.

This Letter proposes and validates a new model for the coefficient of thermal expansion additivity in the binary silicate glasses employed most frequently in fiber optic systems. This simplistic model can be used as a tool to predict the additivity of α for a variety of silicate-based glass compositions. Additionally, it was shown that a good prediction of α in the case of the BaO- SiO_2 system can be successfully used to determine dn/dT for amorphous BaO by removing the α mismatch due to fiber geometry.

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